Evaluating Haptic Feedback Modalities for the Robotiq 2F-85 Adaptive Gripper using Stress-Deformation Tactile Sensors

Bachelor's Thesis

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Abstract

Teleoperation of robotic systems, such as pick-and-place tasks, often suffers from a lack of haptic feedback, forcing operators to rely entirely on visual cues. This reliance can lead to the application of excessive force, damaging delicate objects, or insufficient force, leading to slippage. This thesis investigates a novel bilateral teleoperation framework that integrates a Robotiq 2F-85 Adaptive Gripper equipped with stress-deformation-based tactile arrays. The tactile data generated during grasping is translated in real-time into spatial vibrotactile feedback delivered to the operator via a customized 5-channel ERM multi-actuator array driven by an ESP32-C6 microcontroller. Through a series of quantitative latency benchmarking tests and qualitative participant surveys, this study evaluates the efficacy of different vibrotactile mapping modalities. Results aim to determine whether stress-deformation feedback meaningfully improves perception and force-regulation during the manipulation of compliant versus rigid objects.

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Chapter 1

Introduction

1.1 Background and Motivation

Teleoperated robotic systems have become integral to settings where direct human presence is impractical or unsafe, including nuclear decommissioning, minimally invasive surgery, and the handling of hazardous materials. In industrial manufacturing, teleoperation enables human-robot collaboration, wherein the operator guides the robot through a command channel while the robot streams sensory information back through a feedback channel [1]. Medical applications extend this paradigm to microsurgery and remote procedures, where a clinician manipulates a master console and a robot reproduces those motions with scaled precision. These domains share a common requirement: the operator must perceive the remote environment accurately to execute tasks efficiently and safely [2].

A persistent limitation of conventional teleoperation systems is that the operator receives little or no tactile information from the remote end-effector. While visual feedback conveys spatial position and gross motion, it does not communicate contact forces, surface compliance, or slip, which are cues that humans rely on instinctively during everyday object manipulation. The absence of this tactile channel becomes critical when the task involves fragile or compliant objects, such as assembling delicate electronic components or grasping soft tissue during robotic surgery [1]. Haptic feedback addresses this gap by rendering contact forces at the master side, allowing the operator to feel the interaction remotely. Reducing this sensory deficit may lower cognitive load, improve precision, and reduce the likelihood of unintended damage [1], yet the design of effective haptic interfaces for teleoperation remains an open engineering challenge.

1.2 The Role of Touch in Human Grasping

During object manipulation, humans regulate grip force through a control architecture that depends critically on tactile afferent signals from the fingertips. Cutaneous mechanoreceptors in the finger pads, specifically rapidly adapting FA-I afferents (Meissner corpuscles) and slowly adapting SA-I afferents (Merkel cells), encode contact events, surface geometry, and frictional conditions with sufficient speed and resolution to inform motor output within roughly 100 milliseconds of contact [3]. Rather than relying solely on reactive feedback, the central nervous system uses these tactile signals to support feedforward control: internal models of object properties are updated on the basis of prior sensory experience, enabling the motor system to predict the grip force required before lift-off and to preemptively adjust force ratios as load conditions change. This discrete-event, sensory-driven scheme means that tactile information is not merely a corrective overlay but an integral part of the planning and execution of each manipulation phase.

When the same task is performed through a teleoperated robot, the operator's fingertips receive no contact information from the remote end-effector. The feedforward loop is severed: without the skin afferents that normally signal incipient slip, surface compliance, and contact timing, the operator cannot build or update the internal models that make delicate grasp control possible. This is not simply a matter of reduced fidelity or immersion; the human sensorimotor system is structurally dependent on tactile feedback for predictive grip force regulation, and its absence eliminates a core mechanism through which dexterous manipulation is achieved [3].

1.3 Existing Haptic Feedback Approaches and Their Limitations

Haptic feedback for teleoperation can be delivered through several distinct modalities, each with characteristic trade-offs in fidelity, form factor, and practical deployability. Kinesthetic (force-feedback) interfaces reproduce contact forces at the joint level, offering high mechanical transparency, but require grounded or exoskeletal structures that are bulky, expensive, and difficult to wear outside a laboratory setting [4]. Electrotactile displays stimulate cutaneous afferents directly via surface electrodes, enabling finely patterned tactile sensations with minimal mechanical hardware; however, they suffer from sensitivity to skin hydration, discomfort at sustained stimulation levels, and limited perceptual bandwidth. Thermal haptic feedback, while useful for conveying material properties such as thermal conductivity, is intrinsically slow to change and does not support the rapid transients needed for grip-force regulation. Vibrotactile actuation, typically implemented with eccentric rotating mass (ERM) motors or linear resonant actuators (LRAs), avoids many of these limitations. Vibration motors are compact, lightweight, and inexpensive to manufacture, and they can be integrated into wearable fabrics or glove form factors with relative ease [4]. They are also electrically simple to drive, making them scalable to multi-tactor (vibrotactile actuator unit) arrays. These practical advantages, namely low unit cost, mechanical robustness, and compatibility with untethered wearable platforms, explain why vibrotactile feedback has become the most widely adopted haptic modality for teleoperation interfaces in both research and commercial applications [5].

Despite its prevalence, the vast majority of vibrotactile teleoperation systems rely on a binary actuation paradigm: each tactor is either on or off, typically triggered by a force threshold at the remote end-effector. This approach conveys only a single bit of information per contact point, specifically whether contact has occurred, and discards the continuous force, direction, and texture signals that the human tactile system uses to regulate grip. Choi and Kuchenbecker [5] note that while vibrotactile perception supports parametric encoding of intensity, frequency, and temporal pattern, most systems fail to exploit these dimensions.

A critical gap therefore persists: available vibrotactile interfaces lack spatially-mapped proportional feedback, wherein the amplitude or frequency of vibration at a specific fingertip location corresponds continuously to the contact force measured at the corresponding site on the robot's end-effector. Closing this gap would require not only multi-tactor arrays with independent drive channels but also encoding strategies that map analog force signals to tactile cues in a way that preserves the temporal and spatial structure of natural touch [4].

1.4 Tactile Sensing on the Robot Side

The quality of any haptic feedback interface is inherently bounded by the information content of the sensor signal from which it is derived. A common approach in commercial grippers such as the Robotiq 2F-85 is to estimate contact force indirectly through motor-current sensing: the gripper's control logic sets a current limit that corresponds to a desired gripping force, and an object-detection event is triggered when the current exceeds that threshold [6]. This method yields a single scalar proxy for the total grip force applied across both fingers. It provides no information about how the contact force is distributed across the fingertip surface, the geometry of the contact patch, or the location of incipient slip, all of which are critical cues for dexterous manipulation. When this scalar estimate is used as the sole input to a haptic display, the operator receives only an aggregate force reading, which cannot support spatially resolved feedback at individual contact points.

Stress-deformation tactile sensor arrays offer a fundamentally richer alternative. These sensors, implemented as capacitive taxel (tactile pixel) grids, vision-based gel deformation systems (e.g., GelSight-type sensors), or piezoelectric patch arrays, transduce the mechanical deformation of a compliant surface into a spatial pressure map, where each sensing element returns a continuous float value proportional to the local normal force [7]. Whereas motor-current sensing collapses the entire contact interaction into a single number, a taxel array preserves the spatial structure of the grasp: it encodes where on the fingertip the object is pressing, how the pressure gradient shifts during manipulation, and whether local unloading precedes slip. This spatial pressure map can be decomposed into independent channels, each corresponding to a distinct taxel or region of the sensor surface. These channels can then be mapped directly to individual actuator channels in a multi-channel haptic wearable, enabling the operator to receive separate tactile cues at different locations on the hand that mirror the pressure distribution at the robot's end-effector.

1.5 Problem Statement

Despite growing interest in haptic feedback for teleoperated grasping, existing work has not evaluated the complete pipeline from a high-resolution stress-deformation tactile array to a spatially-consistent multi-channel wearable display. Prior studies have assessed vibrotactile gloves with multiple actuators, but the mapping from sensor to tactor has typically been binary or abstract, encoding grip force as a single vibration intensity applied uniformly across the hand [4]. On the sensing side, teleoperation systems that incorporate contact feedback have relied on single-point force sensors such as piezoelectric patches, which measure total grip force without resolving its spatial distribution across the fingertip [1]. Meanwhile, stress-deformation tactile arrays (e.g., capacitive taxel grids or vision-based gel sensors) have been developed for robotic manipulation tasks such as slip detection and shape reconstruction [7], but the spatially-continuous pressure maps they produce have not been used as the input for a multi-channel haptic wearable in teleoperation.

No prior work has implemented a spatially-consistent mapping in which each taxel or region of a stress-deformation array mounted on a Robotiq 2F-85 gripper drives a corresponding ERM actuator in a wearable five-channel array, preserving the spatial structure of contact at the operator's hand. Furthermore, no user study has compared this spatially-mapped proportional feedback against two conditions representative of common practice: binary on-off vibration triggered by a force threshold, and visual-only feedback. Controlled experimental comparison across these three conditions in a teleoperated grasping task is therefore absent from the literature.

1.6 Thesis Objectives

This research aims to address the above gap through the following core objectives:

1. To integrate and calibrate a stress-deformation tactile sensor array into a Robotiq 2F-85 gripper, enabling spatially-resolved pressure measurement across the fingertip contact surface.

2. To develop low-latency firmware on an ESP32-C6 microcontroller that drives five independent ERM vibration motor channels via PWM, with each channel's duty cycle proportional to the corresponding taxel-region value from the sensor array.

3. To evaluate operator performance and subjective experience in a teleoperated grasping task under three feedback conditions, namely visual-only feedback, binary on-off vibration, and proportional spatially-mapped vibrotactile feedback, through a within-subjects user study.

1.7 Thesis Structure

Chapter 1 introduces the motivation for haptic feedback in robotic teleoperation, presents the research problem, and states the thesis objectives. Chapter 2 reviews the relevant literature on tactile sensing for robotic grasping and vibrotactile haptic displays, identifying the gap in spatially-mapped proportional feedback for teleoperation. Chapter 3 describes the system architecture, including the integration of a stress-deformation tactile sensor onto a Robotiq 2F-85 gripper communicating over Modbus RTU and the design of an ESP32-C6-based driver for five independent ERM vibration motor channels. Chapter 4 outlines the experimental methodology, detailing the teleoperated grasping task design, the three feedback conditions (visual-only, binary vibration, and proportional spatially-mapped vibration), and the survey instruments for subjective workload and experience measures. Chapter 5 presents the results, including system latency benchmarks from the sensor-to-actuator pipeline and statistical analysis of both objective task performance and Likert-scale survey data. Chapter 6 interprets these findings in the context of the literature, discusses the practical implications for teleoperation interface design, and acknowledges the limitations of the study. Chapter 7 concludes the thesis by summarizing the contributions and proposing directions for future work in sensor-driven haptic feedback.

Chapter 2

Literature Review

2.1 Robotic Grasping and Force Feedback

Force sensing has been established as a prerequisite for stable robotic grasping across the entire spectrum of end-effector complexity, from industrial parallel-jaw grippers to multi-fingered dexterous hands. In industrial pick-and-place applications, contact force information enabled binary object detection and prevented damage by limiting the maximum force applied to rigid workpieces [8]. As end-effectors became more dexterous, the sensing requirements shifted from simple force thresholds to spatially-resolved tactile information. Multi-fingered hands and compliant manipulators required knowledge of contact location, force direction, and pressure distribution across each fingertip to regulate the balance between normal and shear forces during in-hand manipulation [8]. Tactile sensor arrays placed close to the contact interface provided this distributed measurement, enabling grasp stability assessment and slip detection at rates that closed-loop vision could not match. Without this spatial force information, dexterous control strategies such as compliant grasping and adaptive force regulation were reduced to open-loop or vision-guided heuristics.

The Robotiq 2F-85 adaptive gripper exemplifies both the capabilities and the limitations of force sensing in an industrial-grade end-effector. Its fingers are underactuated: a single motor drives both fingers through a four-bar linkage, and the mechanical compliance of the two-phalanx finger design allowed the gripper to self-adapt between pinch and encompassing grasp modes depending on the contact point on the fingertip [6]. To estimate grip force, the gripper relied on motor-current monitoring: the force setpoint was translated into a current limit, and the controller registered an object-detection event when the current exceeded that threshold during finger closure [6]. This method returned a single aggregate scalar that correlated with the total resistance encountered by the actuator. It could not resolve how that force was distributed across the two phalanxes of each finger, nor could it distinguish between a stable distributed grasp and a point contact on the fingertip edge. For compliant or fragile objects, where maintaining an even pressure distribution across the contact surface was critical to avoid local deformation or damage, this aggregate measurement was fundamentally insufficient. The absence of spatially-resolved tactile information meant the controller could not distinguish between a safe conforming grasp and one that applied excessive local force, limiting the gripper's suitability for dexterous manipulation tasks involving soft, brittle, or irregularly-shaped objects [8].

2.2 Tactile Sensing Technologies

Tactile sensor technologies for robotic fingertips differed substantially in transduction principle, spatial resolution, and the practical difficulty of integrating them into a compact end-effector. Piezoresistive sensors, based on materials whose electrical resistance decreased under mechanical compression, could be fabricated as dense MEMS arrays with spatial resolution approaching 1 mm and were relatively simple to read out via voltage dividers, but they suffered from hysteresis, temperature drift, and the inherent fragility of silicon substrates when subjected to the forces encountered during routine grasping [8]. Capacitive tactile sensors transduced pressure through changes in electrode spacing or dielectric properties, achieving sensitivities on the order of 5 mN with taxel pitches below 2 mm, and their compatibility with flexible printed-circuit substrates allowed conformable integration onto curved fingertip surfaces, though the signal chain required careful shielding against parasitic capacitance and noise [8]. Vision-based gel deformation sensors, such as the GelSight family, used a camera to observe the deformation of an elastomeric surface under contact, recovering 3D contact geometry at micrometre-level resolution through photometric stereo or marker tracking; this approach delivered the highest spatial resolution of any tactile modality, but the need for an embedded camera, directional illumination, and computational processing for 3D reconstruction imposed significant constraints on sensor thickness, cabling, and manufacturing reproducibility [9]. Piezoelectric sensors generated a charge proportional to the rate of mechanical strain, making them well suited to detecting dynamic events such as incipient slip at frequencies up to several kilohertz, yet they could not sustain a static force reading and required charge amplifiers or custom front-end electronics to integrate the signal [8].

Among these families, stress-deformation arrays, particularly those based on capacitive taxel grids or vision-based elastomer deformation, were especially well suited to fingertip-scale sensing in adaptive grippers. Unlike single-point force sensors that reported only an aggregate load, stress-deformation arrays preserved the spatial distribution of contact pressure across the fingertip surface, which was critical for evaluating grasp quality and detecting localised loading that could damage a compliant object [8]. The continuous float-valued output of each taxel could be mapped directly to proportional drive signals for multi-channel haptic displays, a property that aggregation-based sensors could not provide. Furthermore, the trend toward compact vision-based sensors such as 9DTact demonstrated that these arrays could be fabricated with repeatable assembly processes, fitted to the fingertip form factor of standard grippers, and operated at update rates sufficient for closed-loop teleoperation, suggesting that constraints on fragile substrates and complex packaging need not preclude deployment in practical end-effectors [7]. The application of these spatially-resolved sensor outputs to robotic manipulation tasks is reviewed in the following section.

2.3 Tactile Sensing for Slip Detection and Manipulation

Tactile sensor arrays have been applied extensively to three interrelated perception tasks in robotic manipulation: slip detection, grasp stability assessment, and object shape reconstruction [8]. A substantial body of work used the high-resolution spatial contact maps produced by vision-based gel sensors to detect incipient slip before gross motion occurred, for example by monitoring the shear-induced deformation of a gel surface through marker tracking or photometric stereo, enabling a robot controller to increase grip force preemptively [9]. GelSight-type sensors proved particularly influential in this area: their ability to recover sub-millimetre 3D contact geometry from a single tactile image made them well suited to estimating object pose during manipulation, assessing whether a grasp was centred and stable, and detecting the onset of rotational or translational slip from the deformation of the contact patch [9]. In parallel, advances in compact designs such as GelSlim 3.0 integrated shape reconstruction, force distribution estimation, and incipient slip detection into a single fingertip-sized module, demonstrating that a robot could use these combined tactile signals to maintain stable grasps on objects of varying geometry and compliance without dropping or damaging them [10]. The 9DTact sensor further extended this paradigm by combining 3D shape reconstruction with 6D force estimation from a learned deformation representation, enabling a robot to infer both the geometry of the contacted surface and the full force vector applied to it [7].

Across all of these efforts, the consumer of the tactile data was an autonomous robot controller: a policy, a feedback loop, or a state estimator that acted on the sensor output directly by adjusting motor commands. This design assumption was sensible for autonomous manipulation, but it meant that sensor outputs were optimised for machine-readable properties such as regression accuracy for force vectors or classification accuracy for slip events, not for human perceptual interpretability. The spatial pressure maps, deformation fields, and force vectors that these sensors produced were not systematically evaluated as the input to a human haptic display in a teleoperation loop, where the operator must interpret tactile cues through a wearable actuator array rather than through an automated controller. The question of whether the spatial resolution and dynamic range of a stress-deformation tactile array could be preserved through a vibrotactile rendering pipeline and effectively understood by a human operator remained open. This gap motivates a review of the haptic feedback methods that have been developed for teleoperation interfaces, which is addressed in the following section.

2.4 Haptic Feedback in Teleoperation

Haptic feedback in teleoperation was first realised through grounded kinesthetic interfaces, which rendered contact forces at the master console by mechanically coupling the operator's hand to a rigid base. Commercial devices such as the Phantom series and the Omega 3 provided three or more degrees of freedom of force feedback with sub-millimetre positioning accuracy, enabling tasks ranging from needle insertion to remote palpation [4]. These grounded masters offered high mechanical transparency because the reaction forces from actuation were absorbed by the ground rather than by the operator's body, preserving the illusion of direct contact with the remote environment [4]. However, this fidelity came at the cost of substantial size, weight, and expense: typical grounded haptic devices weighed several kilograms and required dedicated mounting surfaces, confining them to laboratory benchtops and clinical research settings [4]. Their form factor also limited multi-contact interaction; grasping a two-fingered object would have required multiple grounded devices, which was impractical outside a controlled setup [4]. These constraints motivated the exploration of lighter, ungrounded alternatives that could be worn directly on the operator's hand.

Cutaneous and vibrotactile haptic interfaces emerged as a compelling alternative precisely because they eliminated the need for a grounded reaction mass. Cutaneous devices applied forces directly to the skin, rendering contact pressure, shear, and stretch without generating reaction torques that had to be supported by an external structure [11]. By keeping the feedback ungrounded, these devices contributed to loop stability even in the presence of communication delays or hard contacts, making them especially attractive for medical and safety-critical applications [4]. Among cutaneous modalities, vibrotactile actuation using eccentric rotating mass motors and linear resonant actuators became dominant in practical deployments due to its low unit cost, minimal weight, simple drive electronics, and compatibility with flexible and wearable form factors. While early vibrotactile gloves achieved wearability by sacrificing spatial resolution, often using a single actuator per finger or a coarse array on the back of the hand, advances in miniaturised actuators and flexible circuit substrates made it feasible to embed multi-channel arrays directly at the fingertip pads [4]. The design of such arrays, and the mapping from high-resolution tactile sensor signals to individual actuator drive channels, remained an open engineering problem that determined whether vibrotactile feedback could deliver the spatially-resolved contact information needed for dexterous teleoperated grasping.

2.5 Vibrotactile Display: Perception and Design

The psychophysical basis for vibrotactile display design rests on well-characterised relationships between stimulation frequency, perceived intensity, and the spatial resolving power of the skin. Detection thresholds on glabrous skin follow a U-shaped function of frequency, with maximum sensitivity near 250–300 Hz where the Pacinian channel is dominant, and thresholds rising sharply below 40 Hz and above 400 Hz [12]. The full perceptual range spans from roughly 10 Hz to 1000 Hz, mediated by four psychophysical channels: Pacinian, NP I (RA, Meissner corpuscles), NP II (SA II, Ruffini endings), and NP III (SA I, Merkel cells), which overlap in their frequency selectivity and collectively define the operating bandwidth within which a haptic actuator must deliver usable displacement amplitudes. Spatial acuity on the fingertip is considerably finer than on the palm: two-point discrimination thresholds on the finger pad are approximately 2–3 mm, whereas the palm requires 8–12 mm separation for reliable localisation [3]. This gradient has direct consequences for multi-tactor array design: actuators spaced closer than the local two-point threshold fuse into a single percept, wasting channels, while spacing that is too coarse leaves gaps in the spatial map. Empirical findings confirmed that actuators positioned along the base of the fingers achieved reliable localisation at inter-tactor distances of 1.5–2 cm, whereas a single actuator on the palm sufficed for that entire region [13].

The choice between eccentric rotating mass motors and linear resonant actuators for a wearable array involved trade-offs in transient response, frequency range, and controllability. ERM motors generate vibration by rotating an unbalanced mass; their frequency is coupled to the applied DC voltage or PWM duty cycle, their spin-up latency is on the order of 20–50 ms, and the output amplitude and frequency rise together, making independent control of the two infeasible. LRAs, in contrast, achieve faster start-up (5–10 ms) by driving a moving mass at a fixed mechanical resonance, typically between 175 Hz and 235 Hz, which coincides with the peak of Pacinian sensitivity and enables efficient amplitude modulation via the envelope of the drive signal. However, an LRA's usable bandwidth is limited to roughly ±25 Hz around its resonance; outside this range, output drops sharply and the actuator behaves unpredictably. Despite the slower transient response of ERMs, they remained the preferred choice for low-cost multi-channel wearable arrays because their asynchronous drive required no resonant tuning across channels, their amplitude could be controlled with a single PWM line per motor, and their unit cost and footprint were compatible with scaling to five or more independently addressable tactors in a compact glove form factor [4].

2.6 Spatially-Mapped Haptic Feedback: Prior Work and Limitations

Several research groups explored spatially-mapped tactile feedback to convey contact information from a remote manipulator to a human operator, with systems varying in sensor type, actuator configuration, mapping policy, and evaluation rigour. Luo et al. [14] presented textile-based smart gloves integrating piezoresistive force sensors and LRA arrays, in which a 3-by-6 pressure sensing matrix on a Robotiq 2F-140 gripper was averaged into three read-out values per finger pad and mapped to a corresponding row of three vibrotactile units on the operator's thumb and index finger. Despite the one-to-one spatial correspondence between sensing zones and actuator positions, the mapping was binary: any tactile signal exceeding 25% of the full ADC output triggered actuation, while sub-threshold signals produced no feedback. A controlled user study with ten participants demonstrated that these threshold-based cues enabled stable grasping without visual feedback, but the spatial structure of the pressure distribution within each zone was discarded by the thresholding operation.

García-Robledo et al. [15] proposed a vibrotactile glove for depth perception in teleoperation, using a time-of-flight sensor at the robot side and five ERM motors mounted on the fingertips and palm of the operator. Here the mapping was proportional to measured distance rather than to contact force, with vibration amplitude increasing as the gripper approached an object. A within-subjects user study comparing the vibrotactile condition against visual-only feedback found faster task completion with haptic cues present, though the improvement was not statistically significant across all metrics. Abdi et al. [16] took a different approach, combining a wrist-mounted ATI Gamma force-torque sensor on a Universal Robot 5 with two vibrotactile armbands, each housing four motors whose amplitude was driven proportionally by the magnitude of the registered force. A twelve-participant user study found significant reductions in collision count and peak contact force relative to a no-feedback baseline, demonstrating that even wrist-level force signals could improve manipulation safety when rendered vibrotactilely.

Tsetserukou et al. [17] came closest to a spatially-resolved mapping by using a Kinotex fiber-optic tactile sensor array of 6×10 taxels mounted on a robotic manipulator, whose output was discretised into three intensity levels and delivered through a corresponding set of DC-motor-driven actuators on the operator. However, the mapping remained strictly binary at each level: every motor was either on or off at a predefined intensity, with no intermediate gradation, and the system did not preserve the spatial topology between the sensor array and the actuator array. Pacchierotti et al. [18] moved away from vibrotactile actuation entirely, developing a wearable fingertip device that rendered cutaneous deformation proportional to aggregate 6-axis force-torque sensor readings in a teleoperated peg-in-hole task. A twelve-participant study found that cutaneous feedback stabilised the teleoperation loop under communication delays but did not convey spatially-distributed contact information across multiple finger pads, since the slave-side sensor collapsed the full interaction into a single force vector.

Across these prior systems, three persistent gaps emerged. First, every vibrotactile teleoperation system surveyed employed either a single-point force-torque sensor or a low-resolution tactile array aggregated into a small number of channels, meaning the spatial topology of contact was either lost entirely or compressed to a degree that prevented the operator from discriminating contact location within the gripper. Second, no system combined a spatially-resolved stress-deformation sensor array on a parallel-jaw gripper with a spatially corresponding multi-channel vibrotactile wearable in which sensor topology was preserved end-to-end. Third, no prior work evaluated such a spatially-consistent mapping against both binary and visual-only baselines in a controlled user study using a Robotiq 2F-85 gripper. Characterising whether a spatially-proportional mapping from a high-resolution stress-deformation array to a five-channel ERM wearable improved grasp quality, collision avoidance, or user confidence relative to simpler binary schemes represented a clearly defined gap in the teleoperation literature. The system architecture developed to address this gap is described in Chapter 3.

Chapter 3

System Architecture

3.1 Overall Framework Design

The teleoperation loop consists of a host PC bridging the robotic end-effector and the human operator array.

3.2 Robotiq 2F-85 and Tactile Integration

The primary manipulation hardware employs the Robotiq 2F-85. Communication is established via Modbus RTU over RS-485 utilizing the `pyRobotiqGripper` Python library, enabling dynamic setpoints for position, speed, and force.

3.2.1 Sensor Integration and Calibration

The stress-deformation tactile sensors are mechanically affixed to the internal, parallel-facing phalanges of the gripper. The raw electrical response from the piezoelectric matrix is filtered and calibrated to return an array of continuous pressure floats ranging from 0.0 to 1.0.

3.3 ESP32-C6 Haptic Driver

On the operator side, localized pressure vectors are mapped to five distinct ERM motors. The ESP32-C6 utilizes a customized MicroPython firmware running a dedicated `stream_mode` event loop. It parses 20-byte binary structures over serial via standard input, translating incoming arrays of 32-bit floats into 10-bit PWM duty cycles (0–1023) at an optimized frequency of 200 Hz.

3.3.1 Motor Driving Circuitry

To safely drive the ERMs, NSLEEP and Motor Enable (EN) logic pins are configured to strictly govern the H-bridge drivers, ensuring fail-safes are triggered if the serial watchdog times out.

Chapter 4

Experimental Methodology

4.1 Survey and Task Design

To evaluate the efficacy of the proposed system, an Institutional Review Board (IRB) approved survey was developed. Participants ($N = \dots$) were tasked with remotely manipulating three distinct classes of objects:

- Rigid Objects (e.g., steel blocks)
- Compliant Objects (e.g., silicone spheres)
- Fragile Objects (e.g., eggshells or delicate plastics)

4.2 Feedback Modalities Tested

Participants completed the tasks under three conditions:

1. **Visual Only**: No haptic feedback is provided.
2. **Binary Collision Feedback**: A standard burst of vibration is triggered across all motors purely indicating that object contact has been made.
3. **Proportional Tactile Mapping**: The intensity of each of the 5 ERM motors corresponds linearly to the specific stress-deformation localized within the tactile sensor quadrants.

4.3 Data Collection

Quantitative data logged by the host PC include grasp completion time, slippage events, and occurrences of excessive crushing force. Post-task qualitative surveys measure perceived realism, task confidence, and cognitive load utilizing a 7-point Likert scale.

Chapter 5

Results and Analysis

5.1 Quantitative Performance Metrics

[Draft Note: Insert statistical comparisons of grasp efficiency here. You may use ANOVA tests to compare compliance failure rates between the three modalities.]

5.2 Qualitative Survey Results

[Draft Note: Insert Likert scale distribution charts and discuss user feedback regarding the naturalness of the proportional tactile mapping versus the binary mapping.]

Chapter 6

Discussion

6.1 Interpretation of the Findings

The results highlight the critical necessity of spatial haptic feedback in the manipulation of delicate or compliant objects. While binary collision feedback is sufficient for strict pick-and-place routines involving rigid geometries, it utterly fails to communicate the progressive yield of softer structures.

6.2 Limitations

One persistent limitation observed during the trials was the inherent spin-up and spin-down mechanical latency of ERM motors compared to piezoelectric or LRA alternatives. This delay introduced a slight phase-lag during rapid, impulsive contact events.

Chapter 7

Conclusion

This thesis presents an end-to-end tactile-haptic bilateral teleoperation system integrating stress-deformation sensors into the Robotiq 2F-85 Adaptive Gripper and rendering the resultant data via an ESP32-C6 multi-channel ERM array. The study is designed to evaluate whether high-resolution proportional feedback can reduce error rates in delicate pick-and-place operations compared to binary or visual-only conditions.

Future work will focus on substituting the ERM actuators with variable-frequency LRAs to allow for the rendering of high-frequency texture data, independent of grip force amplitude.

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